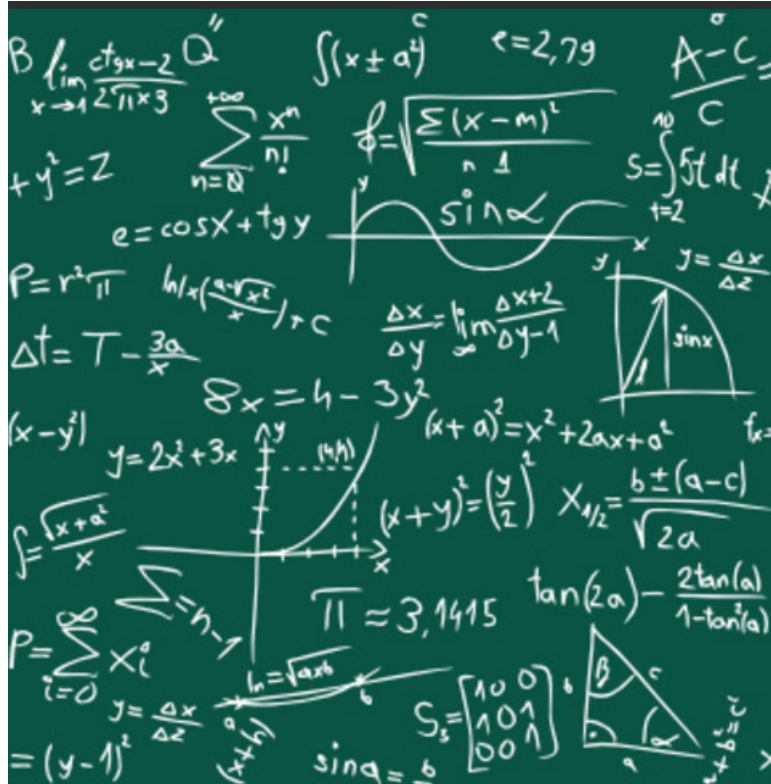


# Tangent lines



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

### Exercise 1

Write the equation of the tangent line to the graph of  $y = f(x)$  at the point  $(x_0, f(x_0))$  for

$$f(x) = 3x^2 + 2x + 1, \quad x_0 = 2$$

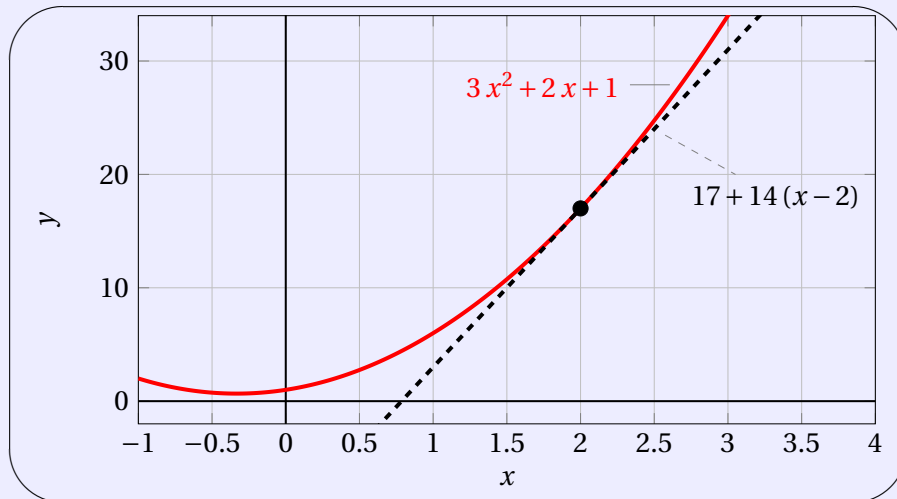
We compute the equation of the tangent line to the graph of the function

$f(x) = 3x^2 + 2x + 1$  at the point with abscissa  $x_0 = 2$ , we have :

$$f(2) = 17, \quad f'(x) = 6x + 2, \quad f'(2) = 14;$$

the tangent line is

$$y = f(2) + f'(2)(x - 2) = 17 + 14(x - 2)$$



### Exercise 2

Write the equation of the tangent line to the graph of  $y = f(x)$  at the point  $(x_0, f(x_0))$  for

$$f(x) = \sin x, \quad x_0 = \frac{\pi}{3}$$

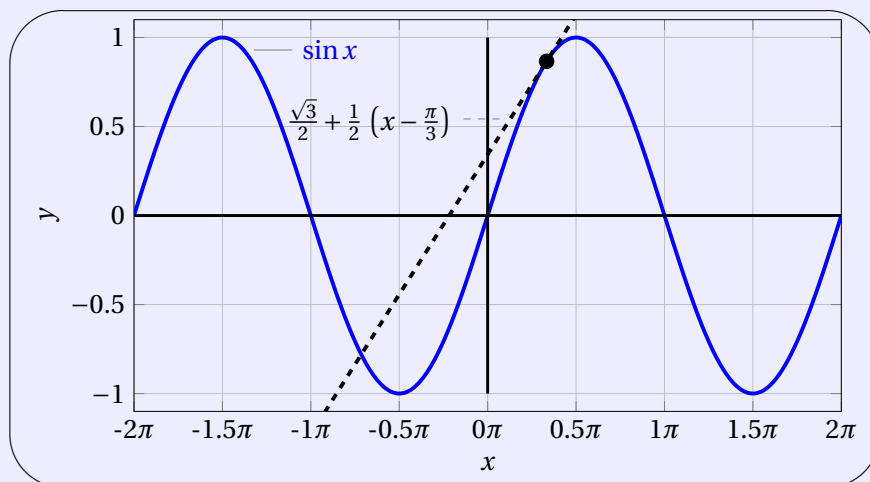
We compute the equation of the tangent line to the graph of the function

$f(x) = \sin x$  at the point with abscissa  $x_0 = \frac{\pi}{3}$ , we have :

$$f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}, \quad f'(x) = \cos x, \quad f'\left(\frac{\pi}{3}\right) = \frac{1}{2};$$

the tangent line is

$$y = f\left(\frac{\pi}{3}\right) + f'\left(\frac{\pi}{3}\right)\left(x - \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} + \frac{1}{2}\left(x - \frac{\pi}{3}\right)$$



### Exercise 3

Write the equation of the tangent line to the graph of  $y = f(x)$  at the point  $(x_0, f(x_0))$  for

$$f(x) = \log x, \quad x_0 = 1$$

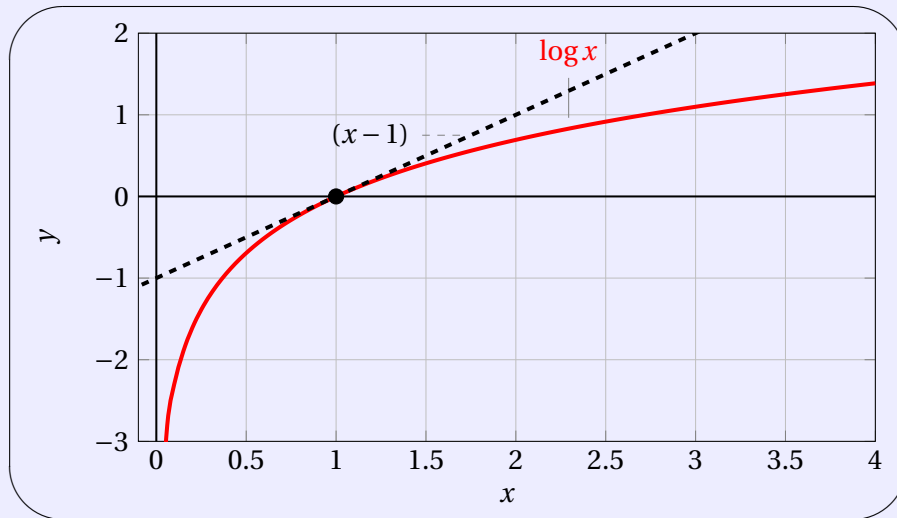
We compute the equation of the tangent line to the graph of the function

$f(x) = \log x$  at the point with abscissa  $x_0 = 1$ , we have :

$$f(1) = 0, \quad f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0} \frac{\log(1 + h)}{h} = 1$$

the tangent line is

$$y = f(1) + f'(1)(x - 1) = 0 + (x - 1)$$



#### Exercise 4

Write the equation of the tangent line to the graph of  $y = f(x)$  at the point  $(x_0, f(x_0))$  for

$$f(x) = a^x, \quad x_0 = 2$$

### Exercise 5

Write the equation of the tangent line to the graph of  $y = f(x)$  at the point  $(x_0, f(x_0))$  for

$$f(x) = (x \log|x|)^3, \quad x_0 = -1$$

### Exercise 6

Write the equation of the tangent line to the graph of  $y = f(x)$  at the point  $(x_0, f(x_0))$  for

$$f(x) = \cos(\log x), \quad x_0 = e^{\frac{\pi}{2}}$$

### Exercise 7

Write the equation of the tangent line to the graph of  $y = f(x)$  at the point  $(x_0, f(x_0))$  for

$$f(x) = e^{x^2}, \quad x_0 = \log 2$$

### Exercise 8

Write the equation of the tangent line to the graph of  $y = f(x)$  at the point  $(x_0, f(x_0))$  for

$$f(x) = e^{-|x|}, \quad x_0 = -1$$